EXP 6

CODE:

```
# Vacuum Cleaner Problem (Simple AI Simulation)


def vacuum_cleaner(start_state):
    # state = (vacuum_position, roomA, roomB)
    # vacuum_position: 'A' or 'B'
    # roomA, roomB: 'dirty' or 'clean'

    state = list(start_state)
    steps = []

    while state[1] == 'dirty' or state[2] == 'dirty':
        pos = state[0]
        if pos == 'A':
            if state[1] == 'dirty':
                state[1] = 'clean'
                steps.append("Clean room A")
            else:
                state[0] = 'B'
                steps.append("Move to room B")
        else:  # vacuum in B
            if state[2] == 'dirty':
                state[2] = 'clean'
                steps.append("Clean room B")
            else:
                state[0] = 'A'
                steps.append("Move to room A")
```

```
    return steps


# Example run
start = ('A','dirty','dirty')
solution = vacuum_cleaner(start)


print("Initial State:", start)
print("Steps to goal:")
for s in solution:
    print("-", s)
```

OUTPUT:

```
============ RESTART: C:\Users\arunp\OneDrive\Documents\AI\EXP 6.py ============
Initial State: ('A', 'dirty', 'dirty')
Steps to goal:
- Clean room A
- Move to room B
- Clean room B
```